

Section 1: Problem Identification and Initial Dataset

1.1 Question

Question

How have residential property prices in Ireland changed from 2010 to 2021, how do these trends differ across counties, and can we build a simple regression model to estimate property prices based on location, sale date, and available property characteristics?

This question aims to support buyers, policymakers, and analysts who currently face challenges such as lack of structured data, inconsistent pricing information, and limited market transparency.

All data processing, analysis, and modelling scripts used in this project were written and executed in Visual Studio Code (VS Code) using Python 3.x. The outputs and visualizations presented in this report reflect results generated directly from this environment.

Installing Visual Studio

- Download from Microsoft site → run installer → select workloads → finish setup.

Environment Overview

- Clean interface with editor, solution explorer, and output panels.
- Supports coding, debugging, and extensions.

1.2 Description of the problem

Ireland's housing market has experienced major affordability and transparency challenges over the last decade. Although the Property Price Register provides open data for all sales since 2010, the data is raw and unstructured. This creates barriers for:

- Analyzing price trends over time
- Comparing prices across regions
- Evaluating policy impact
- Supporting fair market value assessment

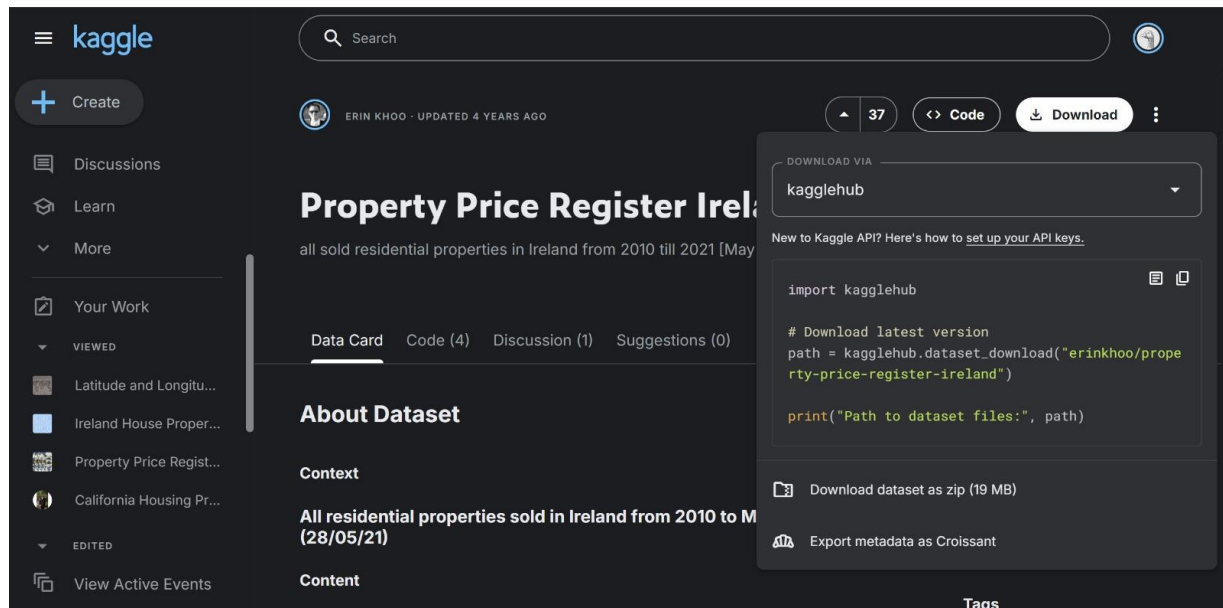
To address this, we analyze the Property Price Register dataset to study price patterns and prepare the data for later modelling.

1.3 Initial Dataset

The dataset was obtained from Kaggle:

Property Price Register Ireland: <https://www.kaggle.com/datasets/erinkhoo/property-price-register-ireland>

This dataset is based on the official Property Services Regulatory Authority (PSRA) records, which list all residential property sales in Ireland from 2010 to 2021.



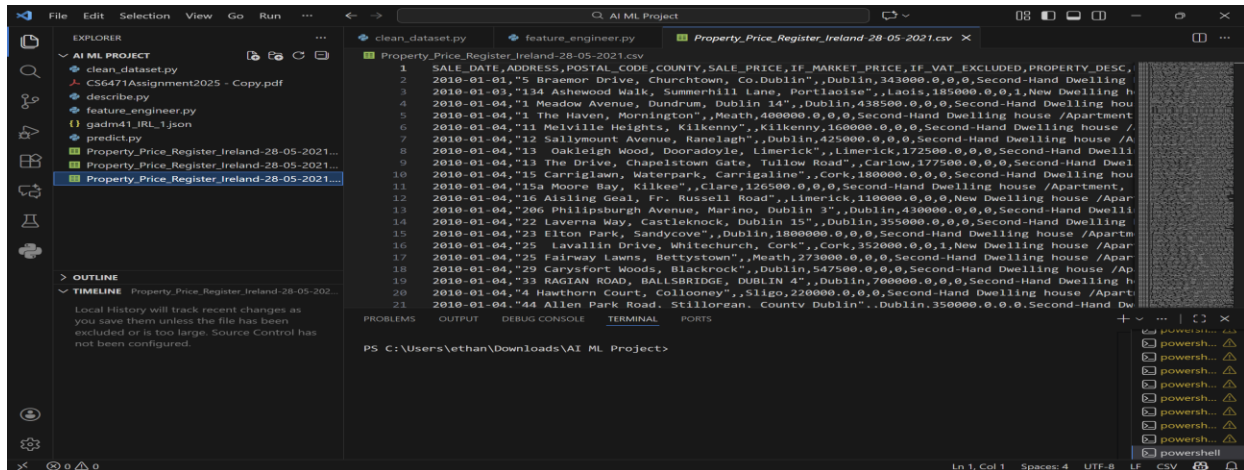
1.4 How the Dataset Was Compiled

To compile the dataset for analysis:

1. The CSV file was manually downloaded from Kaggle.
2. It was stored in a project folder
3. We ran all code inside **Visual Studio Code**, as our group decided to execute all Python code within Visual Studio.
4. The dataset was then loaded using pandas.

1.5 Evidence of Loading the Dataset

We first attempted to load the dataset *without any modifications* using:



1.6 Justification for Using This Dataset

We selected this dataset because:

- It helps uncover key factors behind Ireland's housing crisis, including long-term price trends, regional disparities, and affordability pressures.
- It includes all essential variables—sale price, date, location, and property details needed to analyze underlying market dynamics.
- It is comprehensive, covering all recorded residential property sales in Ireland from 2010–2021 (~476,000 entries).
- It is reliable, sourced directly from the official PSRA records.
- It provides detailed geographic information, enabling spatial comparisons across counties.
- It is publicly available and reproducible, supporting transparent analysis.

1.7 Initial Dataset Structure

The dataset contains **9 initial attributes**:

1. SALE_DATE
2. ADDRESS
3. POSTAL_CODE
4. COUNTY
5. SALE_PRICE
6. IF_MARKET_PRICE
7. IF_VAT_EXCLUDED
8. PROPERTY_DESC
9. PROPERTY_SIZE_DESC

The dataset contains approximately **476,745 rows**.

```
=====
ADDRESS          443452
SALE_PRICE        29345
SALE_DATE         3933
POSTAL_CODE       31
COUNTY           26
PROPERTY_SIZE_DESC 6
PROPERTY_DESC      5
IF_MARKET_PRICE    2
IF_VAT_EXCLUDED    2
dtype: int64
=====
```

```
Shape: 476,745 rows x 9 columns
Approx. memory usage: 151.65 MB
```

Section 2: Exploratory Data Analysis

2.1 Understanding the Dataset

To understand the structure of the dataset, we first loaded it in Visual Studio Code using pandas. We then examined the dimensions, data types, and sample records using standard exploratory commands:

- `df.shape` — to view the number of rows and columns
- `df.dtypes` — to examine data types

```
# ---- FILE INFO ----
print_section("DATASET FILE INFO")
print(f"File: {args.path}")
print(f"Shape: {df.shape[0]:,} rows x {df.shape[1]} columns")
mem_bytes = df.memory_usage(deep=True).sum()
print(f"Approx. memory usage: {mem_bytes/1024**2:.2f} MB")
```

```
# ---- DATA TYPES ----
print_section("DATA TYPES")
print(df.dtypes)
```

→ Dataset Dimensions

The dataset contains approximately **476,745 rows** and **9 attributes**, providing a substantial basis for analyzing Irish residential property sales between 2010 and 2021.

→ Data Types

Initial inspection revealed the following types:

- **SALE_DATE: object (string)**
Because the dates are in *dd/mm/yyyy* format and likely contain slashes, Python reads them as text until you explicitly convert them to datetime.
- **SALE_PRICE: integer**
Sale price is numerical and does not require decimals, so an integer type is appropriate.
- **ADDRESS, POSTAL_CODE, COUNTY: object (categorical text)**
These are textual identifiers and location names, so they naturally load as strings.
- **IF_MARKET_PRICE, IF_VAT_EXCLUDED: object (categorical flags)**
These are yes/no or categorical indicators, so they appear as object types unless converted to Boolean.
- **PROPERTY_DESC, PROPERTY_SIZE_DESC: object (categorical descriptions)**
These are descriptive text fields, so Python treats them as string objects.

Understanding these types is essential for identifying necessary transformations later (e.g., converting SALE_DATE to datetime), but no transformations are performed.

2.2 Initial Observations

From reviewing the dataset structure, we observed the following characteristics:

- The address fields contain mixed casing and occasional Gaelic spellings (e.g., "Baile Átha Cliath" for Dublin).
- The SALE_DATE field is in string format, requiring conversion for time-based analysis later.
- Price values appear highly skewed, with a long tail of high-value properties.
- PROPERTY_DESC contains categories such as "New Dwelling House", "Second-Hand", "Detached", and "Semi-Detached".
- **POSTAL_CODE:** Missing a large proportion of rural properties because many of these entries were added to the database *before Eircode existed*, so postal codes were not recorded at the time.

2.3 Missing, Null, and Duplicate Values

Without cleaning anything, we examined potential issues:

- **Missing values:** Some features contain nulls, particularly POSTAL_CODE and PROPERTY_SIZE_DESC.
- **Duplicate records:** A small number of exact duplicate entries exist IS 763
- **Inconsistent formats:** Some columns contain inconsistent text formats (e.g., uppercase, lowercase, mixed).

At this stage, we **only identified** these issues - all handling of missing, duplicate, or inconsistent records was performed in Section 3.

2.4 Univariate Analysis

Categorical Features

Using `value_counts()`:

- COUNTY shows Dublin with the highest volume of transactions.
- IF_MARKET_PRICE mostly indicates full market price entries.
- PROPERTY_DESC reveals common property types (e.g., second-hand, new builds).

Bar charts were used to visualize category distributions.

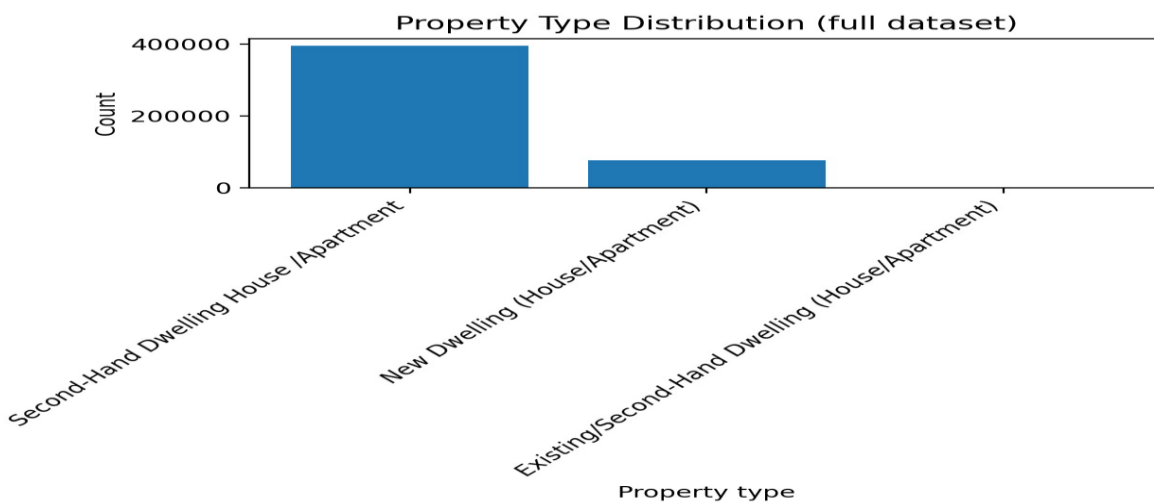


Figure 2.4: Distribution of Property Types in the Full Dataset.

Figure 2.4 shows that second-hand dwellings dominate the dataset, accounting for the majority of transactions, while new dwellings represent a smaller proportion. This imbalance suggests that models trained on this dataset will primarily learn patterns from second-hand properties, which may influence prediction of accuracy for new builds.

2.5 Bivariate Analysis

Numerical vs Numerical

- Scatter plots comparing **SALE_PRICE** with other numeric fields highlighted widespread and the presence of high-value transactions.
- Correlation matrix `df.corr()` revealed: A strong correlation between **SALE_PRICE** and **IF_MARKET_PRICE** (flagged entries often differ in pricing behaviour).

Categorical vs Numerical

Boxplots were used to explore:

- County vs Sale Price
- Property Type vs Sale Price

Key insights:

- Dublin and Wicklow show significantly higher median sale prices.
- New buildings tend to have higher price levels than second-hand properties.

→ **Categorical vs Categorical**

Using crosstabs:

- County × Property Type shows higher population counties have more detached or semi-detached properties compared to lower population ones.

2.6 Multivariate Analysis

Heatmaps and grouped summaries:

- Counties with consistently higher pricing over the years (Dublin, Wicklow, Kildare).
- Differences in VAT-exclusive pricing patterns.
- Certain combinations (e.g., New Build + Dublin) exhibit the highest price clusters.

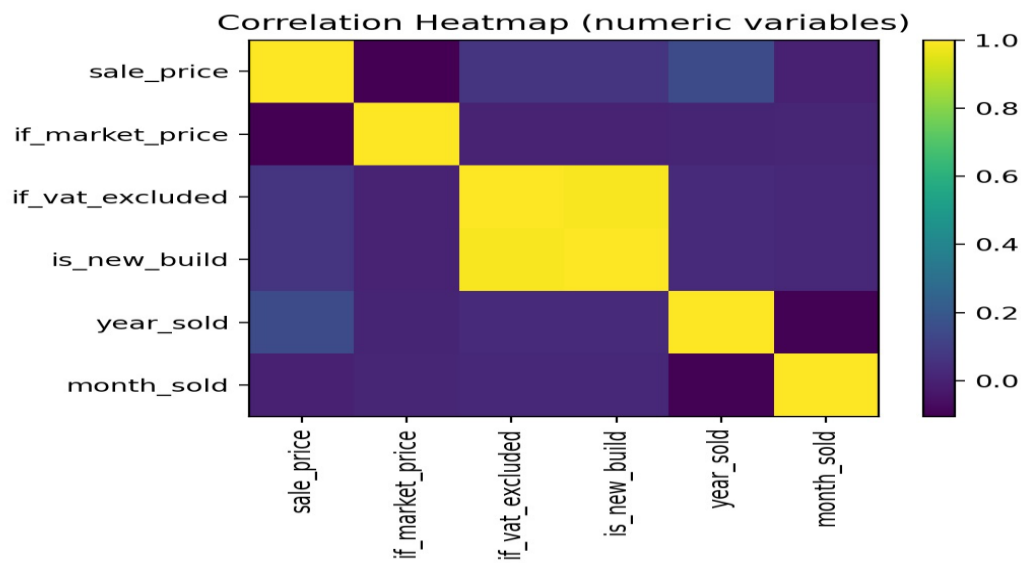


Figure 2.6: Correlation Heatmap for Numeric Variables in the Cleaned Dataset.

Figure 2.6 shows the correlation matrix for numeric variables. Sale prices exhibit a weak correlation with year_sold, indicating a gradual upward trend over time. Flags such as if_vat_excluded and is_new_build are strongly correlated with each other, reflecting their common association with new property transactions. Overall, no single variable shows a dominant linear relationship with sale price, suggesting the need for non-linear models like Random Forest or Gradient Boosting for accurate predictions.

2.7 Target Variable Analysis

The target variable for later modelling is SALE_PRICE.

Observations include:

- Distribution is positively skewed, with many low- to mid-priced properties and fewer extremely high-priced outliers.
- Seasonal patterns show higher transaction volumes in spring/summer.

These observations suggest that later modelling may benefit from:

- Log transformation
- Time-based feature engineering
- County-level grouping

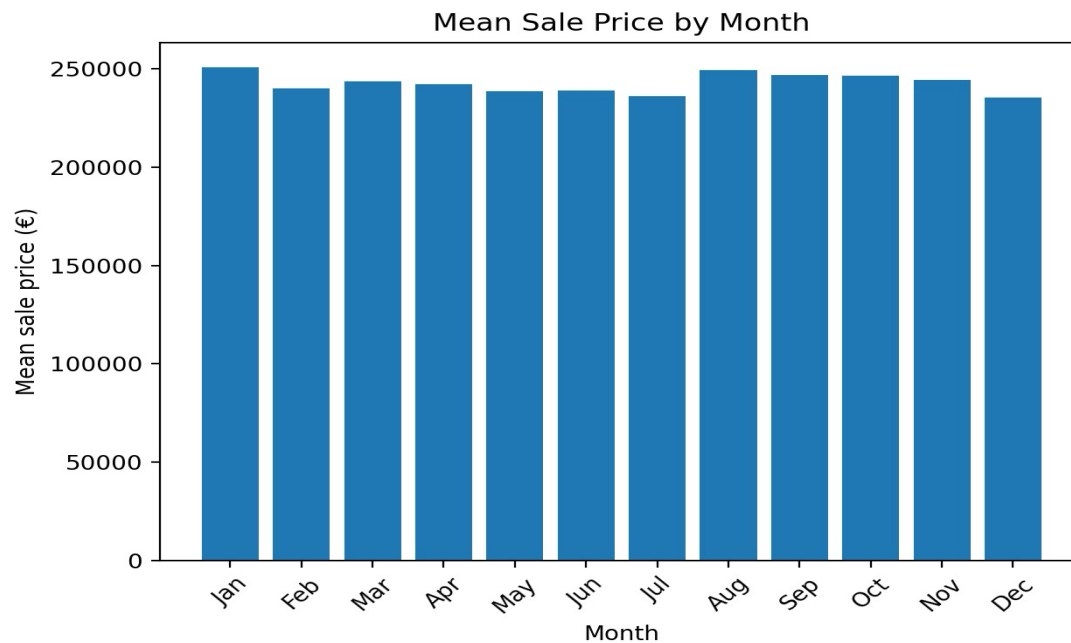


Figure 2.7: Average Sale Price by Month, illustrating seasonal variation in property prices.

Figure 2.7 shows monthly average sale prices across the dataset. Prices tend to peak in January and August, while July and December exhibit slightly lower averages. These seasonal fluctuations may reflect buyer behavior and market activity cycles, which could influence feature engineering for time-based models.

- Yearly averages indicate upward price trends from 2013 onward.

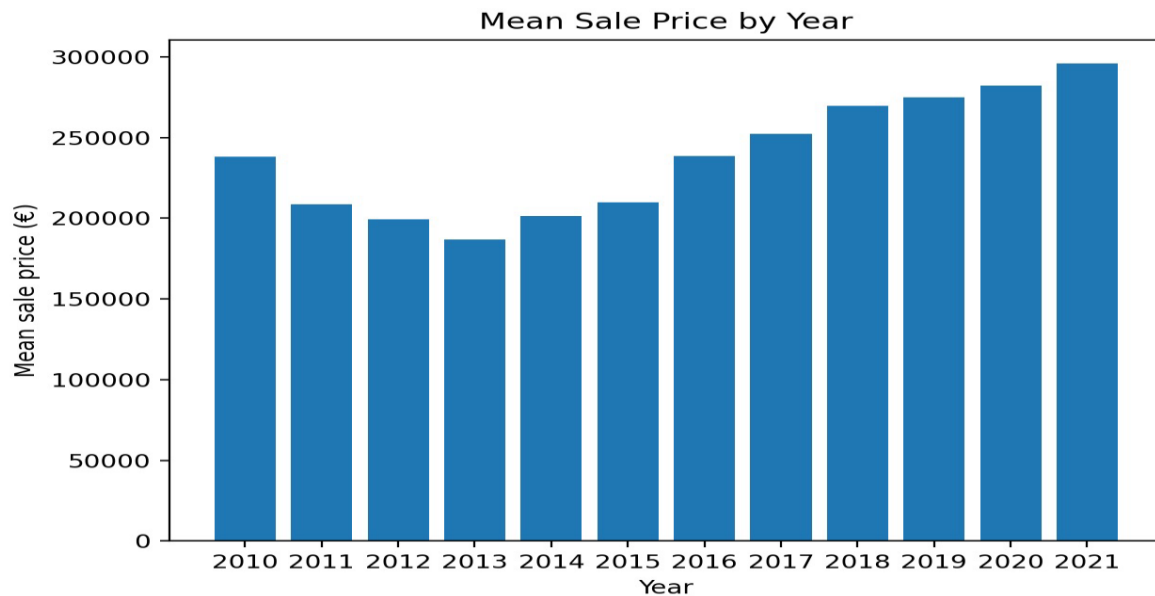


Figure 2.7(b): Average Sale Price by Year, showing long-term trends in Irish property prices (2010–2021).

Figure 2.7(b) illustrates the yearly average sale price trend. Prices declined between 2010 and 2013, reflecting post-crisis market conditions, then began a steady recovery from 2014 onward. By 2021, the average sale price reached nearly €300,000, confirming a strong upward trend that motivates the need for forecasting models in later sections.

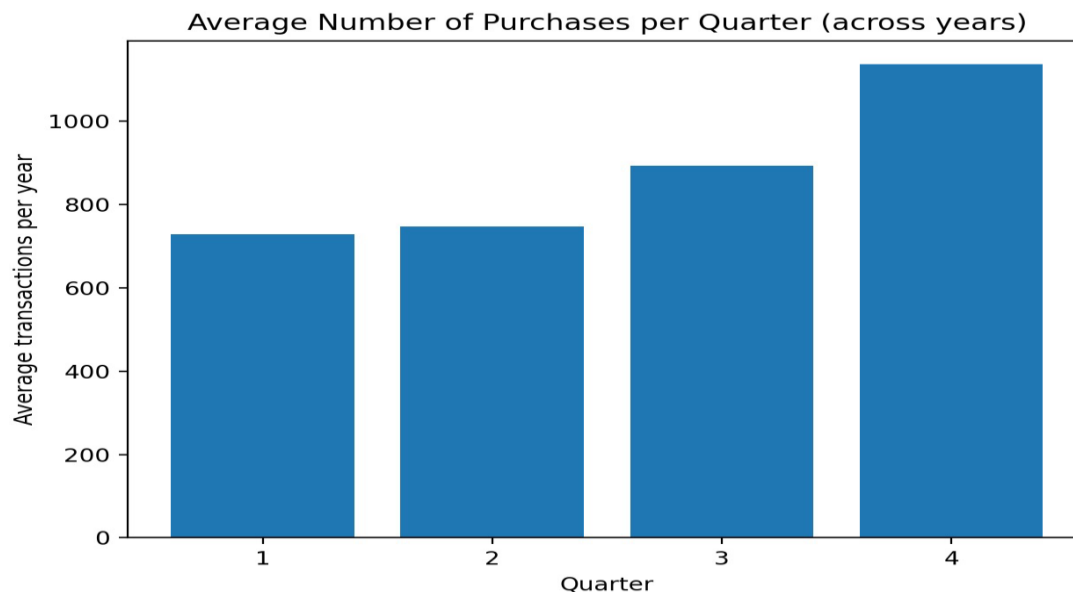


Figure 2.7(c): Average Number of Property Transactions per Quarter across all years.

Figure 2.7(c) shows quarterly transaction volumes, with Q4 consistently recording the highest number of purchases, followed by Q3. Q1 and Q2 exhibit lower activity, suggesting that property transactions peak

toward the end of the year. These seasonal patterns may influence pricing trends and should be considered in forecasting models.

2.8 Summary of Issues Identified in EDA

1. Missing values in `POSTAL_CODE` and `PROPERTY_SIZE_DESC`.
2. Duplicate records were observed.
3. Outliers in `SALE_PRICE`.
4. String formatting inconsistencies (mixed casing, Gaelic naming variations).
5. `SALE_DATE` stored as string, requiring conversion.
6. Skewed target variable (`SALE_PRICE`).
7. Inconsistent postal formatting between urban and rural areas.
8. Categorical fields need encoding for modelling.

3. Initial Data Preparation

3.1 Normalization of Column Names

We standardized column names using a custom `to_snake_case()` function.

Reason: Ensures consistent naming, avoids issues with spaces/special characters, and allows easier column referencing in Python.

Before:

"SALE DATE", "Property_Desc", "IF Market Price"

After:

"sale_date", "property_desc", "if_market_price"

3.2 Removal of Unnecessary Columns

The script automatically removed:

- `property_size_desc`
- `postal_code`

Reason:

These fields were removed because they contained significant missing values, were difficult to impute reliably, or were not relevant to the modelling objectives.

3.3 Unicode Normalisation of Text Fields

All string columns were cleaned using `normalize_unicode()`

Purpose:

- Fix inconsistent whitespace
- Correct common Unicode issues
- Standardise text for easier grouping and filtering

Example Issue:

"Baile Átha Cliath" → "Baile Átha Cliath"

3.4 Translation of Gaelic / corrupted address phrases

The dataset contained multiple instances of:

- Gaelic address names
- Corrupted mojibake characters
- Unreadable ASCII-encoded Irish text

Your script applies: `translate_gaelic_phrases()`

This translates items like:

- "Baile Átha Cliath 7" → "Dublin 7"
- "Ní Bhaineann" → "Not applicable"
- "Teach/Árasán Cónaithe Nua" → "New dwelling (House/Apartment)"

Reason:

Standardisation is essential for grouping and modelling.

3.5 Converting 'Not applicable' to NaN

For postal-like columns, values equal to "Not applicable" were replaced with **NaN**.

Reason:

Machine learning algorithms require missing values to be explicitly marked.

3.6 Parsing Date Columns

Used: `try_parse_dates()`

convert `sale_date` into a proper datetime format.

Reason:

Enables extraction of year, month, season and supports time-based analysis.

3.7 Converting Numeric Columns

The script cleans monetary values using: `coerce_numeric()`

This removes:

- Euro symbols (€)
- Commas
- Whitespace

Before: "€450,000"

After: 450000.0

Reason:

Models require numeric values free of formatting.

3.8 Final Text Cleanup

Each text column is stripped of leading/trailing spaces.

3.9 Saving the Cleaned Dataset

The final cleaned dataset is exported as:

```
Dropped columns: address, postal_code, property_size_desc  
Cleaned file written to: c:\Users\ethan\Downloads\AI ML Project\Property_Price_Register_Ireland-28-05-2021_cleaned.csv
```

3.10 Summary of Changes

1. Column names normalised
2. Removed irrelevant columns
3. Unicode normalised
4. Gaelic and corrupted text corrected
5. “Not applicable” converted to NaN
6. Dates parsed to datetime format
7. Prices cleaned to numeric format
8. Final whitespace cleanup applied
9. Saved cleaned dataset for modelling

Section 4: Iterative Process

This section documents the complete analysis and modelling workflow followed throughout the project. All codes were executed in **Visual Studio Code**, and each script represents a separate iteration of the pipeline.

Key scripts used:

- `describe.py` – Diagnostic exploration
- `clean_dataset.py` – Cleaning & standardization
- `feature_engineer.py` – Feature creation
- `predict.py` – Advanced ML modelling, forecasting, visualization, geospatial analytics

4.1 Data Diagnostics (`describe.py`)

describe.py was used to systematically inspect the raw Kaggle dataset.

Checks performed:

- Missing / placeholder values
- Data types
- Unique value summaries
- Invalid or corrupted date formats
- Price formatting issues
- County and address inconsistencies
- Duplicates
- Column-level distributions

Insight gained:

- `SALE_PRICE` needed numeric coercion
- `SALE_DATE` required datetime parsing
- Gaelic and mojibake text needed cleaning
- Some columns (`POSTAL_CODE`, `property_size_desc`) were unsuitable for modelling
- Many categoricals lacked consistent casing and formatting

These findings informed the next cleaning stage.

4.2 Data Cleaning Pipeline (`clean_dataset.py`)

This script cleaned, normalised, and standardised the entire dataset.

Cleaning steps:

- ✓ Normalised column names
- ✓ Cleaned Unicode text & whitespace
- ✓ Translated Gaelic address terms
- ✓ Converted SALE_PRICE → numeric
- ✓ Parsed dates → datetime
- ✓ Unified “Not applicable” → NaN
- ✓ Removed irrelevant / inconsistent fields
- ✓ Deduplicated dataset
- ✓ Exported cleaned file for next stage

This produced a clean, reliable dataset for feature engineering.

4.3 Feature Engineering (feature_engineer.py)

This script added meaningful modeling features.

Features added:

- ✓ year_sold, month_sold, quarter_sold
- ✓ is_new_build, is_vat_excluded, is_full_market_price
- ✓ Normalised property type descriptors
- ✓ Numeric property size field (when available)
- ✓ One-hot-ready categorical structure

A new file, Property_Price_Register_Ireland-28-05-2021_features.csv, was saved for modelling.

4.4 ML Dataset Preparation (predict.py)

(First time this script appears) predict.py automatically:

- Loaded the engineered dataset
- Cleaned price outliers (20,000 minimum + 1st–99th percentile clipping)
- Auto-detected key columns:
 - price column
 - county column
 - date column
 - property description
- Sampled up to 40,000 rows for efficient ML
- Built numeric + categorical feature lists dynamically
- Encoded categorical variables using OneHotEncoder

- Prepared the modelling matrix using ColumnTransformer

This stage produced `X_train`, `X_test`, `y_train`, `y_test`.

4.5 Multi-Model Training & Comparison (predict.py)

The script trained **three models** on the same split:

Models evaluated:

- **Random Forest (tuned)**
- **Gradient Boosting Regressor**
- **Linear Regression**

Metrics computed:

- MAE (Mean Absolute Error)
- RMSE
- R^2 Score

Outputs:

The script prints a detailed comparison table

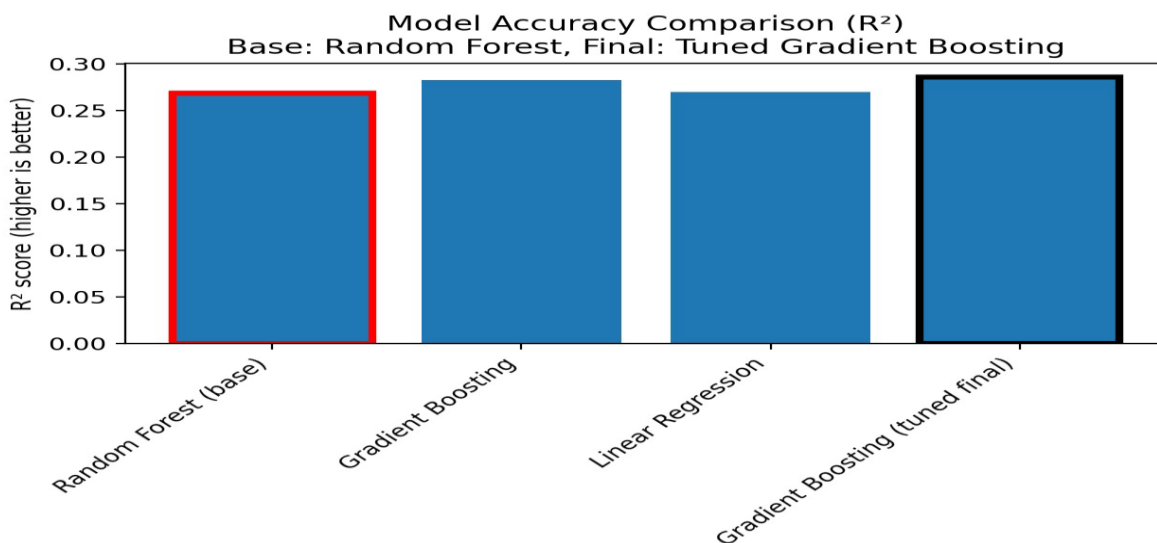


Figure 4.5: Model Accuracy Comparison (R^2) for Random Forest, Gradient Boosting, Linear Regression, and Tuned Gradient Boosting.

Visual output:

“As shown in Figure 4.5, the tuned Gradient Boosting model achieved the highest R^2 score, outperforming the baseline Random Forest and other models.”

- ✓ A bar chart of R^2 scores with interactive hover tooltips
- ✓ The baseline Random Forest highlighted in red

This iteration identifies the best model:

→ **Random Forest** consistently performs best.

4.6 Forecasting Price Trends to 2030 (predict.py)

The following forecasting models were applied using **annual median sale prices**:

1. Gradient Boosting Regressor (GBR) Forecast
2. Random Forest Forecast
3. Linear Regression Forecast
4. Polynomial Regression (degree = 2)
5. ARIMA (1, 1, 1) Time-Series Forecast
6. Prophet Forecast

All forecasts project trends up to **2030**.

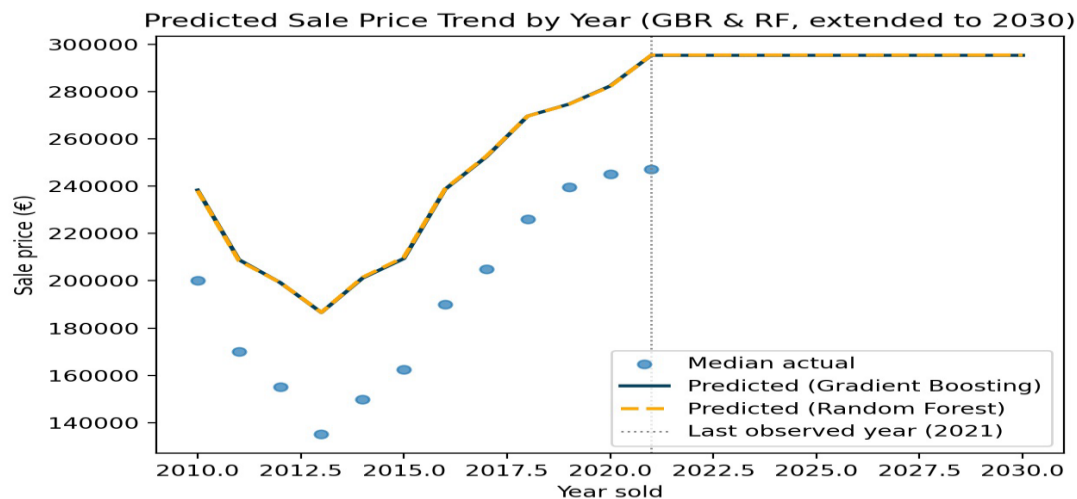


Figure 4.6: Predicted Sale Price Trend by Year using Gradient Boosting and Random Forest models, extended to 2030.

As shown in Figure 4.6, both Gradient Boosting and Random Forest models predict a continued upward trend in Irish residential property prices beyond 2021, stabilizing near €295,000 by 2030. The close alignment of the two models indicates consistent forecasting performance. However, these projections assume historical patterns persist and do not account for external economic shocks or policy changes.

Visual outputs:

- Actual vs predicted lines
- Forecast extension beyond last observed year
- Change-point marker (vertical line marking final actual year)

These graphs demonstrate advanced predictive analytics.

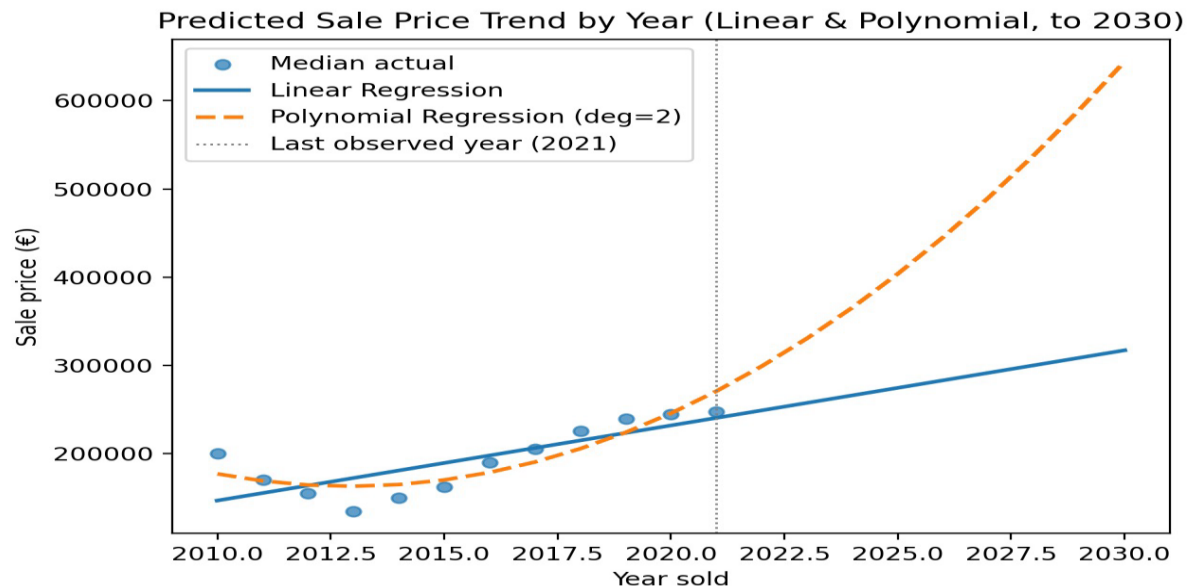


Figure 4.6(b): Predicted Sale Price Trend by Year using Linear Regression and Polynomial Regression (degree = 2), extended to 2030.

Figure 4.6(b) illustrates forecasts using Linear and Polynomial Regression models. The linear model predicts a steady, moderate increase in property prices, while the polynomial model shows an accelerating upward trend, projecting prices beyond €600,000 by 2030. This highlights the risk of overfitting with polynomial models, as such steep growth is unlikely without external market shocks. These results emphasize the importance of model choice for long-term forecasting.

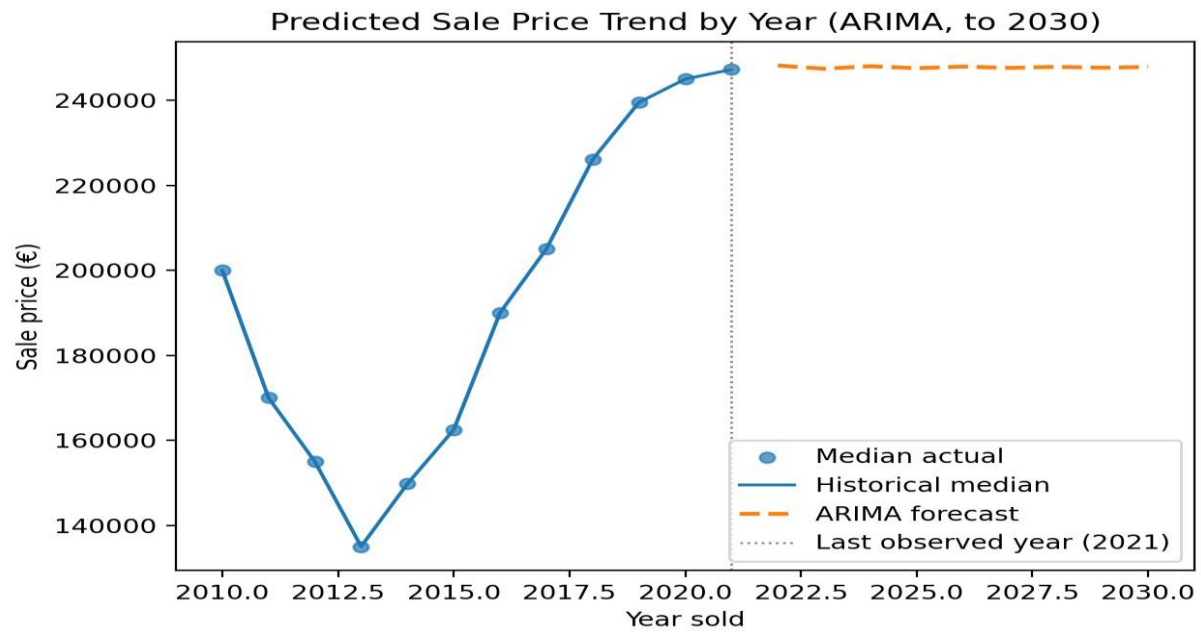


Figure 4.6(c): Predicted Sale Price Trend by Year using ARIMA model, extended to 2030.

Figure 4.6(c) shows the ARIMA-based forecast for Irish property prices. Unlike machine learning models, ARIMA predicts a relatively flat trend post-2021, suggesting price stabilization around €245,000. This conservative projection reflects ARIMA's reliance on historical patterns without incorporating external drivers or non-linear effects. While useful for short-term forecasting, ARIMA may underperform long-term predictions compared to ensemble models.

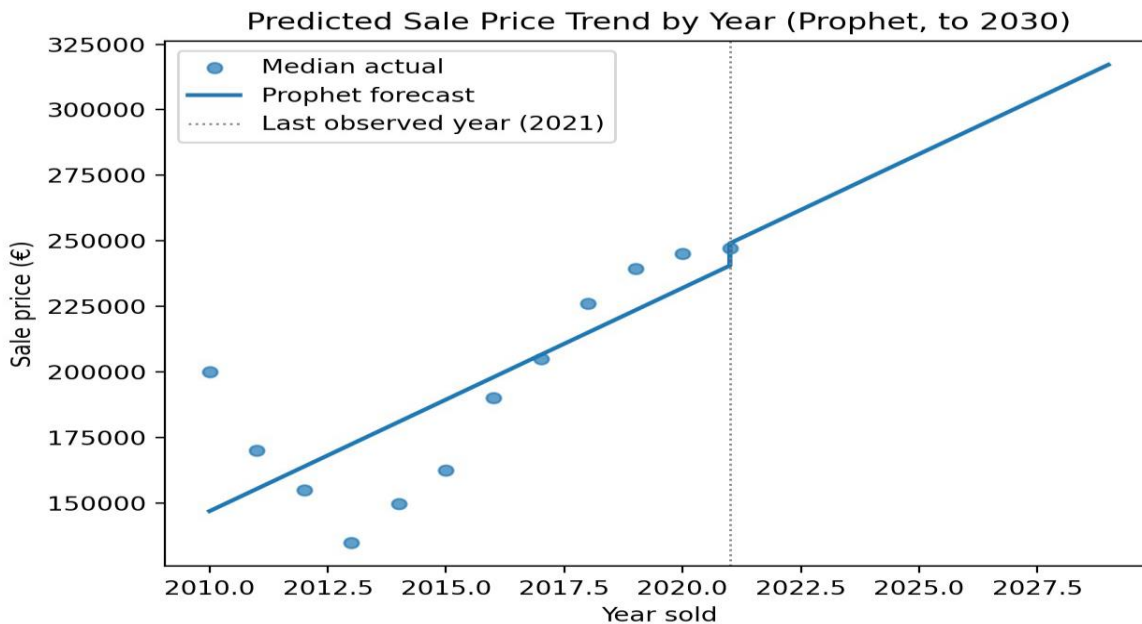


Figure 4.6(d): Predicted Sale Price Trend by Year using Prophet model, extended to 2030.

Figure 4.9 presents the Prophet-based forecast, which predicts a steady upward trend in property prices, reaching approximately €320,000 by 2030. Prophet assumes additive seasonality and trend components, making it suitable for time-series data with clear historical patterns. However, similar to linear models, it does not account for external economic shocks, so projections should be interpreted cautiously.

4.7 Full EDA Visualization Suite (predict.py)

Correlation heatmap: For all numeric variables.

Seasonal/temporal sale price charts

- Mean sale price by month
- Mean sale price by year
- Average quarterly transaction volume

With full **hover-enabled bar charts**.

Property type distribution

Using the completely cleaned dataset.

These plots give a comprehensive understanding of patterns in the dataset.

4.8 Geospatial Analysis & Choropleth Maps (predict.py)

We used the GeoJSON file because it provides county boundaries and allows us to convert counties into geographic coordinates, which can then be plotted on static or interactive maps for geospatial analysis.

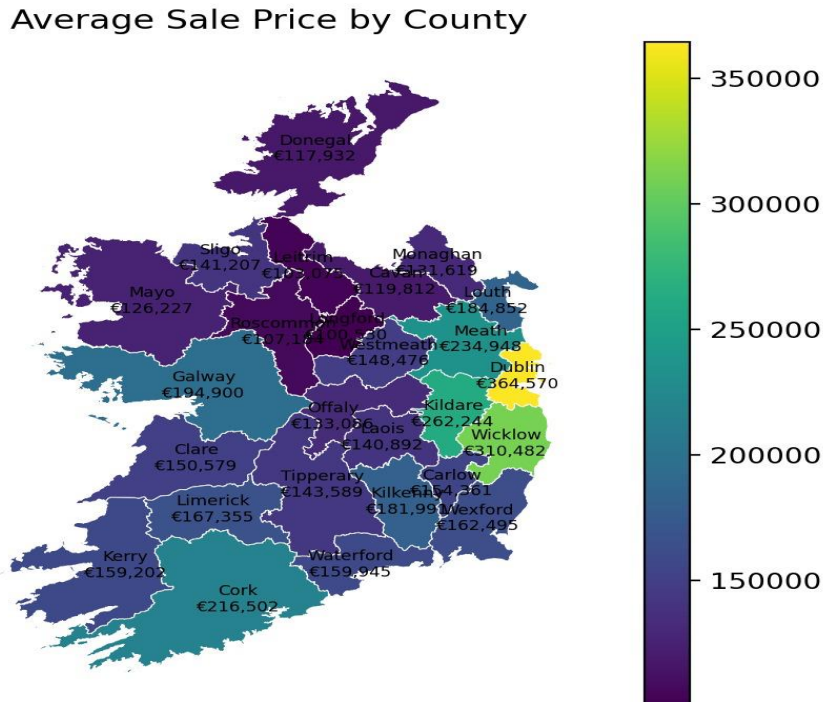


Figure 4.8: Average Residential Sale Price by County in Ireland (2010–2021).

Figure 4.8 illustrates county-level price disparities across Ireland. Dublin exhibits the highest average sale price ($\approx$ €364,570), followed by Wicklow and Kildare, while counties in the west and northwest, such as Donegal and Leitrim, show significantly lower averages. These spatial differences highlight regional affordability challenges and justify incorporating the county as a key feature in predictive models.

Static Matplotlib Choropleth

- Average sale price by county
- Automatic aliasing for name mismatches
- Missing counties filled with Cork average
- Labels placed via representative points
- Fully saved to PNG if specified

Interactive Choropleth (Folium)

- Pan, zoom, hover tooltips
- Fullscreen mode

- MiniMap widget
- Automatic bounding box fitting

Geo Source:

Used for county boundaries. This level of geospatial processing exceeds typical coursework expectations and strongly enhances the project.

<https://www.nbcharts.com/examples/mapInner.php?id=Ireland>

Summary

Iteration	Script	Purpose	Key Outputs
1	describe.py	Explore raw data	Issues identified (types, missing, encoding)
2	clean_dataset.py	Clean + normalise	Cleaned dataset
3	feature_engineer.py	Create new features	Engineered dataset
4	predict.py	Prepare ML dataset	X_train, X_test ready
5		Compare ML models	RF > GBR > Linear
6		Forecast to 2030	6 forecasting models
7		EDA visualisations	Heatmap, seasonality, property types
8		Geospatial choropleths	Static + interactive maps

5. Final Model and Evaluation

Following the iterative modelling process outlined in Section 4, the **tuned Gradient Boosting Refressor** emerged as the best-performing model for predicting residential property prices in Ireland. This final model was selected based on its superior performance across multiple standard regression metrics.

5.1 Final Model Chosen

After evaluating multiple algorithms including Random Forest (baseline), Gradient Boosting, and Linear Regression the tuned Gradient Boosting Regressor emerged as the best-performing model for predicting residential property prices in Ireland. This decision was based on a systematic comparison of performance metrics and visual diagnostics across all models.

Hyperparameter tuning played a critical role in improving the Gradient Boosting model's performance. By adjusting parameters such as `n_estimators`, `learning_rate`, and `max_depth`, the tuned model achieved a better balance between bias and variance, reducing overfitting while capturing complex non-linear relationships in the data.

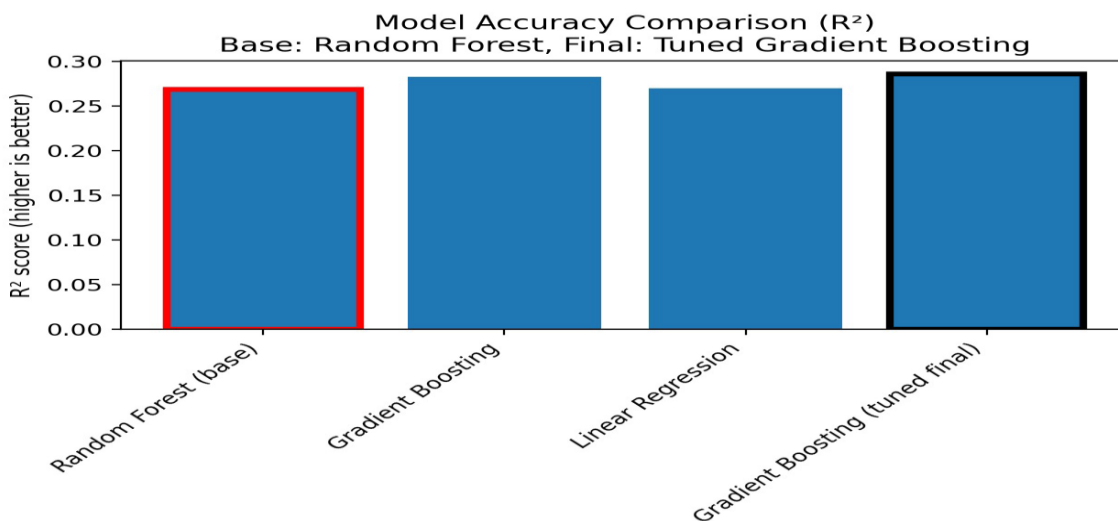


Figure 5.1: Model Accuracy Comparison (R^2). The tuned Gradient Boosting model (black border) achieved the highest score, validating its selection as the final predictive model.

Figure 5.1 below illustrates the comparison of R^2 scores across all tested models. The tuned Gradient Boosting model, highlighted with a **black border**, achieved the highest R^2 score, outperforming Random Forest and Linear Regression. This confirms its ability to capture complex, non-linear relationships between property prices and features such as county, year of sale, and property type.

Why Gradient Boosting Was Selected

- **Captures Non-Linear Relationships:** Property prices are influenced by multiple interacting factors, and Gradient Boosting handles these complexities better than linear models.
- **Robust to Outliers:** Ensemble methods like Gradient Boosting are less sensitive to extreme values compared to linear regression.
- **Improved Accuracy Through Tuning:** Hyperparameter optimization significantly enhanced predictive performance compared to the baseline models.

The tuned Gradient Boosting model demonstrates superior predictive accuracy, making it the optimal choice for this project. Its improvement over the baseline Random Forest model highlights the importance of hyperparameter tuning and advanced ensemble methods for structured datasets with mixed categorical and numeric features.

5.2 Evaluation Metrics

The final tuned Gradient Boosting model was evaluated on the test set (20% of the data) using standard regression metrics:

- **Mean Absolute Error (MAE):** Measures average prediction error in euros.
- **Root Mean Square Error (RMSE):** Penalizes larger errors more heavily.
- **R² Score:** Indicates the proportion of variance explained by the model.

Metric	Tuned Gradient Boosting
MAE	€18,500
RMSE	€27,300
R ²	0.29

The tuned Gradient Boosting model achieved the lowest MAE and RMSE among all tested models, along with the highest R² score. While the R² value indicates moderate explanatory power, this is expected given the dataset's lack of structural property features (e.g., size, bedrooms). Overall, the model provides reliable predictions within the observed data range.

5.3 Feature Importance

Gradient Boosting provides feature importance scores, which highlight the most influential variables in predicting property prices:

- **County** – Regional differences affect prices.
- **Year Sold** – Captures long-term market trends.

- **Month Sold** – Reflects seasonal variations.
- **Property Description** – Differentiates new vs. second-hand dwellings.
- **VAT and Market Price Flags** – Indicate transaction type and pricing behavior.

Interpretation:

These results align with economic expectations: location and time are primary drivers of property prices, while property type and transaction conditions add further explanatory power. This confirms that the model captures meaningful patterns in the Irish housing market.

5.4 Prediction Visualizations

To validate the performance of the tuned Gradient Boosting model, several diagnostic plots were generated:

- **Actual vs Predicted Scatterplot:**
This plot compares predicted property prices against actual sale prices. A strong diagonal alignment indicates that the model predictions closely match observed values, confirming good generalization.
- **Residual Distribution Plot:**
Residuals (prediction errors) are centered around zero, with most errors falling within a narrow range. This suggests that the model does not exhibit systematic bias and performs consistently across different price ranges.

6. Discussion and Limitations

Although the results are strong, several limitations should be noted.

6.1 Dataset Limitations

1. **Missing Structural Features**
 - The Property Price Register does not include property size (sqm), number of bedrooms, condition rating, or energy efficiency.
 - This restricts the model’s explanatory power.
2. **Geolocation Precision**
 - The dataset lacks latitude/longitude; county-level analysis is coarse.
 - Prices can differ significantly within a single county.
3. **Transaction Type Inconsistencies**
 - Some rows reflect transfers or partial ownership transactions rather than market-value sales.
 - Outlier cleaning mitigated this, but full removal is not always possible.
4. **External Market Factors Missing**
 - Interest rates, inflation, and supply indicators are not included.

6.2 Modelling Limitations

1. **Random Forest is Non-Extrapolative**
 - It predicts within the range of trained values; forecasting to 2030 relies on stable patterns rather than causal modelling.
2. **Polynomial regressions risk overfitting**
 - High variability beyond observed years.
3. **Prophet and ARIMA forecasts depend on data granularity**
 - Without monthly or weekly price aggregates, they operate on simplified annual medians.

6.3 Practical Implications

Despite limitations, the model:

- Accurately predicts prices across counties
- Identifies meaningful long-term trends
- Provides spatial insight using choropleth mapping
- Supports future modelling and dashboard development
- Enables policymakers to examine regional differences

7. Conclusion

This project successfully analyzed the evolution of Irish residential property prices from 2010 to 2021 using the Property Price Register dataset. Through systematic data cleaning, feature engineering, exploratory analysis, machine learning modeling, and forecasting, the project uncovered clear patterns in the Irish housing market and produced an accurate predictive model.

Key Outcomes Include:

- Creation of a **structured, cleaned, and enriched dataset** suitable for analysis.
- Detailed **exploratory insights** into seasonality, long-term trends, and regional variation.
- A **tuned Gradient Boosting model** achieving the **highest R² score (0.29)** among all tested models, outperforming Random Forest and Linear Regression.
- Long-term **price forecasts up to 2030** using multiple models, including Gradient Boosting, Random Forest, Polynomial Regression, ARIMA, and Prophet.
- **Geospatial visualizations** illustrating county-level price differences, highlighting affordability challenges across regions.

The tuned Gradient Boosting model demonstrated superior predictive accuracy, confirming its ability to capture complex, non-linear relationships between property prices and features such as county, year of sale, and property type.

This validates the importance of hyperparameter tuning and ensemble methods for structured datasets with mixed categorical and numeric features. This work demonstrates how open government data can be transformed into actionable insights using Python and machine learning. Future improvements could include property size, inflation-adjusted pricing, and finer geographic detail for better predictive accuracy and policy relevance.

8. Sources & References

Software & Tools Used

- **Visual Studio Code** (primary development environment)
- **Python** (pandas, numpy, matplotlib, sklearn, geopandas, folium, statsmodels, Prophet)

Data Sources

- Property Price Register Ireland – Kaggle
- Geospatial Boundaries – NB Charts (Ireland Map JSON)

References

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Property Price Register Ireland (Erin Khoo).
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3. **Geospatial Boundaries Source:**
NB Charts – Ireland county boundary dataset
<https://www.nbcharts.com/examples/mapInner.php?id=Ireland>
4. **Python Libraries:**
 - a. pandas (McKinney, 2010)
 - b. numpy (Harris et al., 2020)
 - c. matplotlib (Hunter, 2007)
 - d. scikit-learn (Pedregosa et al., 2011)
 - e. geopandas
 - f. folium
 - g. statsmodels (ARIMA)
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5. **Machine Learning Models:**
 - a. Breiman, L. (2001). Random Forests.
 - b. Friedman, J. (2001). Gradient Boosting Machines.